Friday Talk

L. DiCarlo et al. **"Demonstration of two-qubit algorithms with a superconducting quantum processor".** In: *Nature* 460.7252 (2009), pp. 240–244

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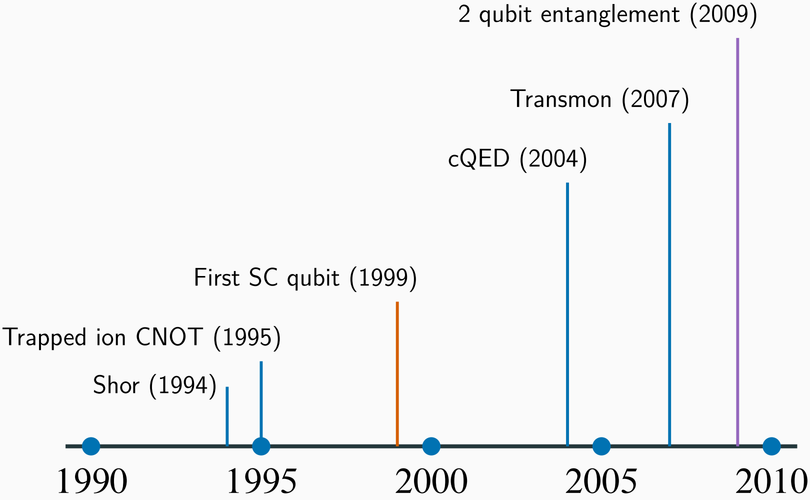
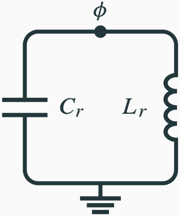
May 8, 2025

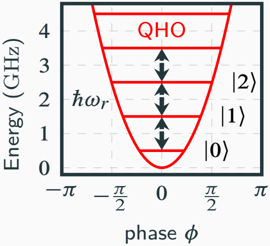
Overview



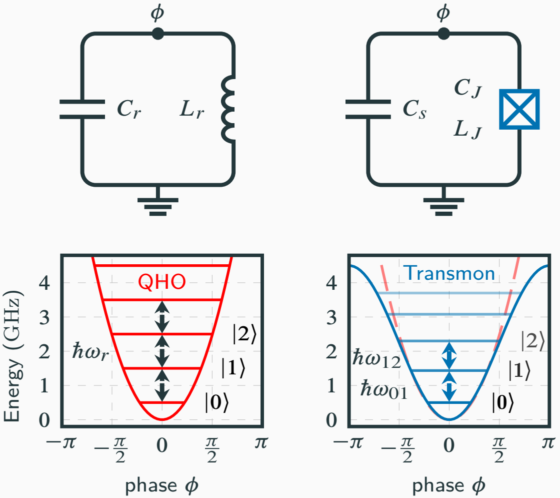
2 qubit entanglement (2009)
Transmon (2007)
cQED (2004)
First SC qubit (1999)
Trapped ion CNOT (1995)
Shor (1994)
1990
1995
2000
2005
2010

Superconducting circuits

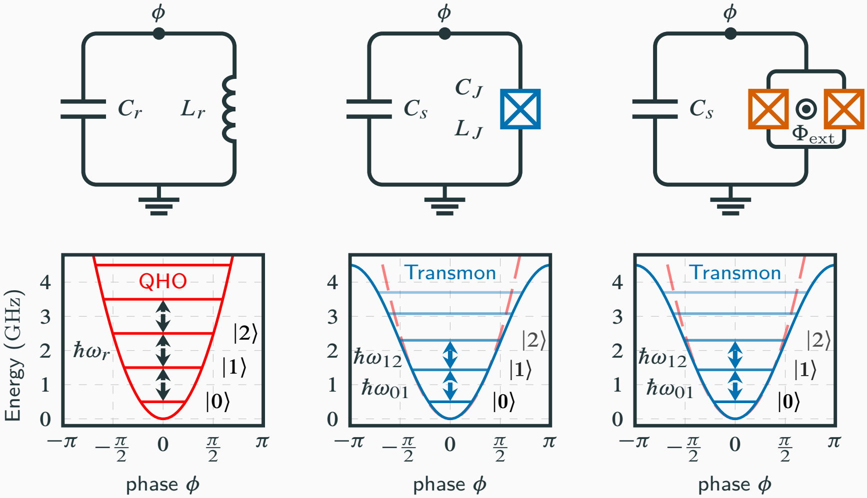




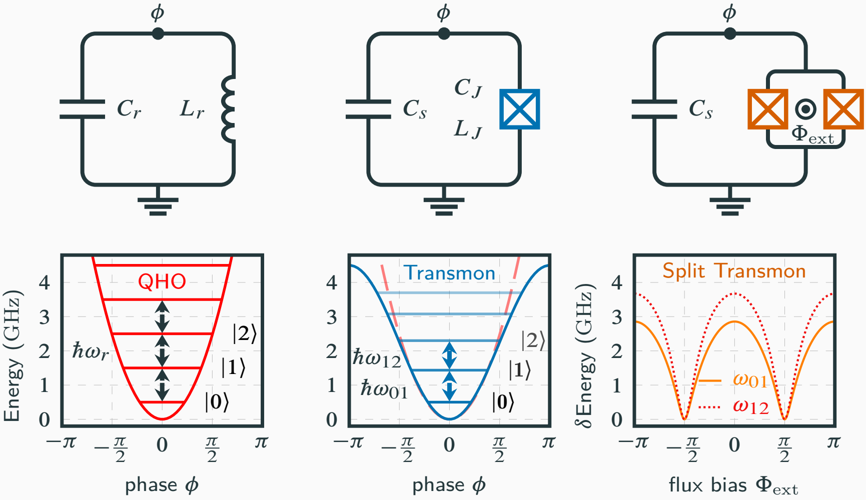
Superconducting circuits



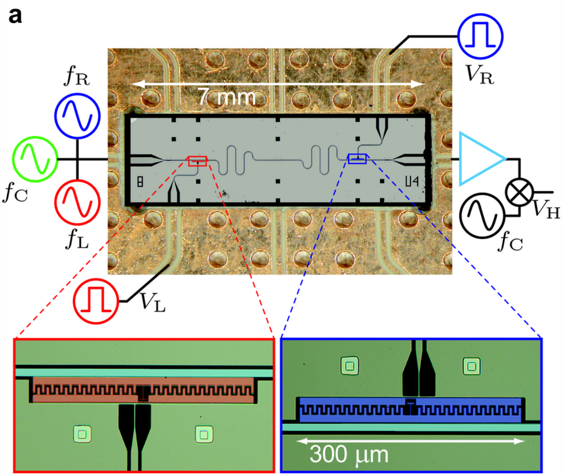
Superconducting circuits



Superconducting circuits



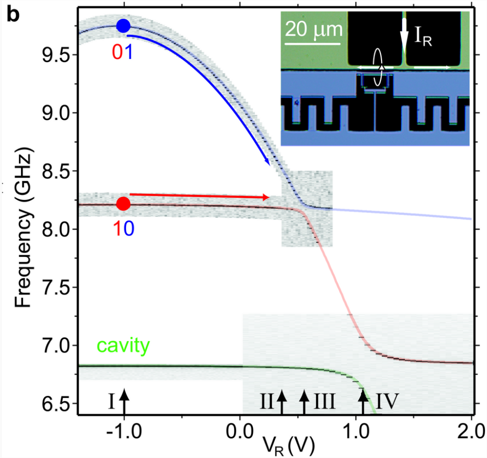
Die shot



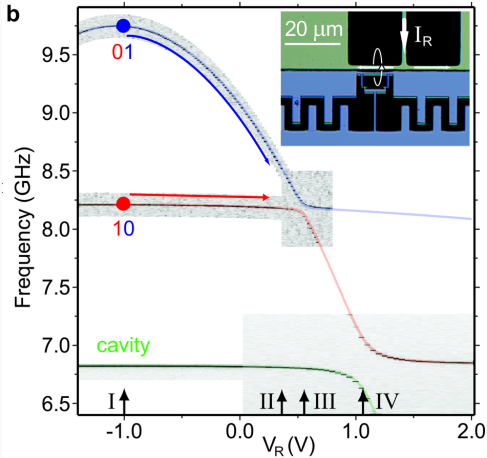
$T_1 = 0.8\ \mu s$ $T_1 = 1.3\ \mu s$

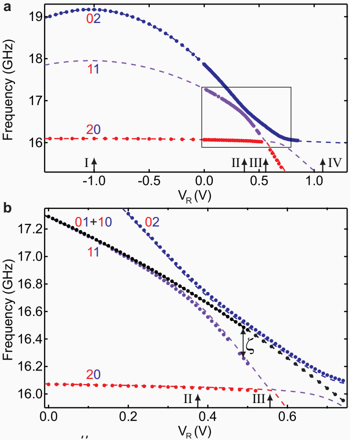
$T_2^* = 1.2\ \mu s$ $T_2^* = 1.8\ \mu s$

Qubit spectrum

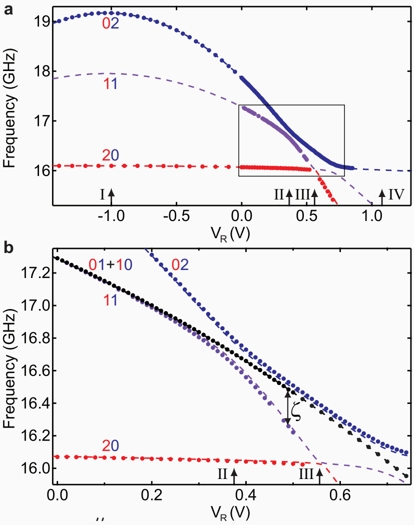


Qubit spectrum





CPhase gate

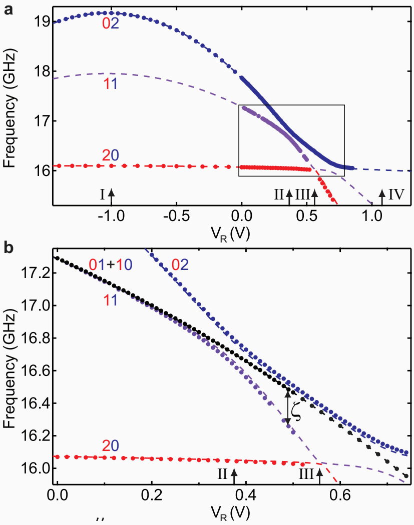


$$U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\phi_{11}} \end{pmatrix}$$

$$\phi_{lr} = 2\pi \int \delta f_{lr}(t)\, \mathrm{d}t$$

$$\phi_{11} = -\int \zeta(t)\, \mathrm{d}t$$

CPhase gate



$$U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{i\phi_{01}} & 0 & 0 \\ 0 & 0 & e^{i\phi_{10}} & 0 \\ 0 & 0 & 0 & e^{i\phi_{11}} \end{pmatrix}$$

$$\phi_{lr} = 2\pi \int \delta f_{lr}(t)\,\mathrm{d}t$$

$$\phi_{11} = \phi_{01} + \phi_{10} - \int \zeta(t)\,\mathrm{d}t$$

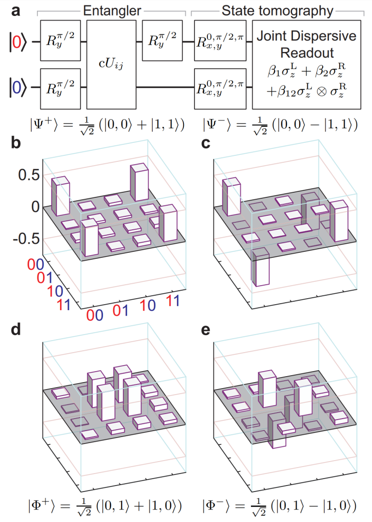
CPhase gate

$$\boxed{cU_{ij}\,|l,r\rangle = (-1)^{\delta_{il}\delta_{jr}}\,|l,r\rangle}$$

	$\boldsymbol{\phi}_{01}$	$\boldsymbol{\phi}_{10}$	$\boldsymbol{\phi}_{11}$
U_{00}	π	π	π
U_{01}	π	0	0
U_{10}	0	π	0
U_{11}	0	0	π

$$U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{i\boldsymbol{\phi}_{01}} & 0 & 0 \\ 0 & 0 & e^{i\boldsymbol{\phi}_{10}} & 0 \\ 0 & 0 & 0 & e^{i\boldsymbol{\phi}_{11}} \end{pmatrix}$$

Qubit entangling



$N = 450000$

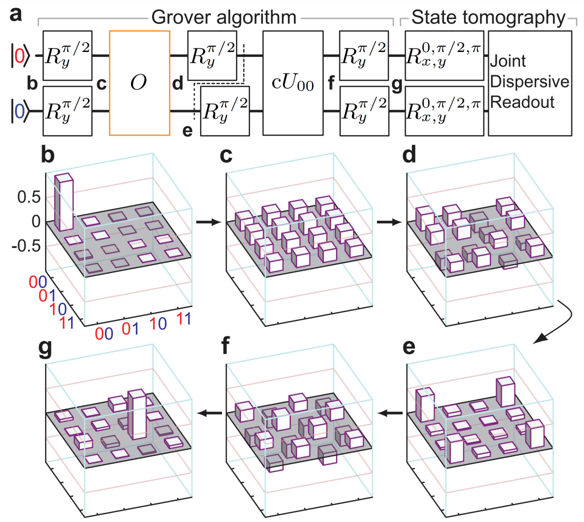
$F = 0.90(1)$

$F = 0.91(1)$

$F = 0.87(2)$

$F = 0.94(1)$

Grover algorithm

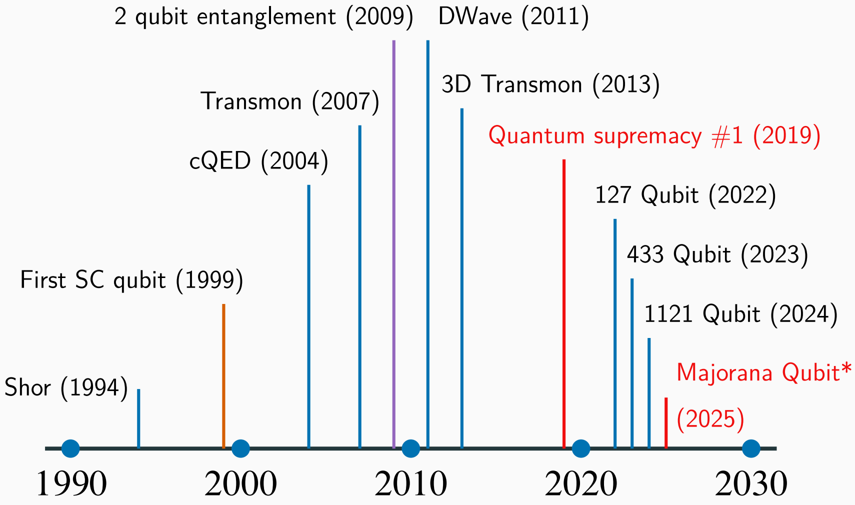


Results

Element		Grover search oracle[a]				Deutsch–Jozsa function[b]			
		f_{00}	f_{01}	f_{10}	f_{11}	f_0	f_1	f_2	f_3
$\langle 0,0\|\rho\|0,0\rangle$	Ideal	1	0	0	0	0	0	1	1
	Measured	0.81(1)	0.08(1)	0.07(2)	0.065(7)	0.010(3)	0.014(5)	0.909(6)	0.841(9)
$\langle 0,1\|\rho\|0,1\rangle$	Ideal	0	1	0	0	0	0	0	0
	Measured	0.066(7)	0.802(9)	0.05(1)	0.054(8)	0.012(4)	0.008(4)	0.031(8)	0.04(2)
$\langle 1,0\|\rho\|1,0\rangle$	Ideal	0	0	1	0	1	1	0	0
	Measured	0.08(1)	0.05(1)	0.82(2)	0.07(1)	0.93(1)	0.93(1)	0.05(1)	0.04(1)
$\langle 1,1\|\rho\|1,1\rangle$	Ideal	0	0	0	1	0	0	0	0
	Measured	0.05(2)	0.07(1)	0.06(1)	0.81(1)	0.05(1)	0.04(1)	0.012(9)	0.07(2)

[a] Uncertainties are based on 10 repetitions. [b] Uncertainties are based on 8 repetitions.

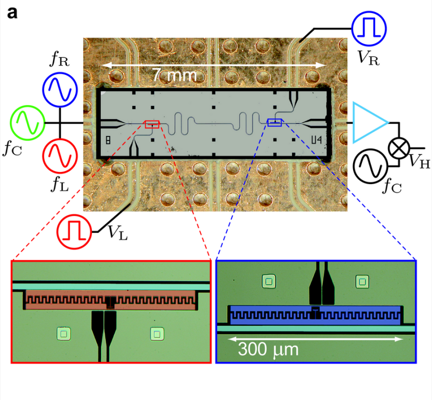
Outlook


2 qubit entanglement (2009)
DWave (2011)
3D Transmon (2013)
Transmon (2007)
Quantum supremacy #1 (2019)
cQED (2004)
127 Qubit (2022)
433 Qubit (2023)
First SC qubit (1999)
1121 Qubit (2024)
Majorana Qubit*
(2025)
Shor (1994)
1990
2000
2010
2020
2030

Questions?

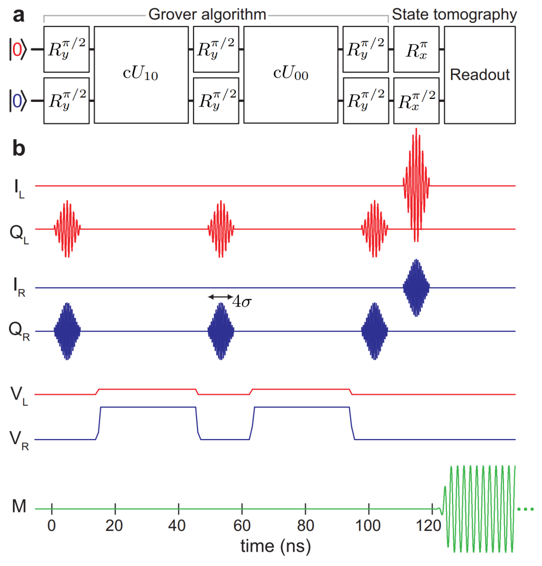
Qubit readout

$$H = \hbar\omega_r a^\dagger a + \frac{\hbar\omega_q}{2}\sigma_z + \hbar\chi a^\dagger a\sigma_z \qquad t(\omega_p) = \frac{\kappa}{\kappa + 2i(\omega_r^{\text{eff}} - \omega_p)}$$



$$\begin{aligned} s(t) &= A_s \cos(\omega t + \phi_s) \\ \mathrm{LO}(t) &= A_{\mathrm{LO}} \cos(\omega t + \phi_{\mathrm{LO}}) \\ V(t) &= s(t)\mathrm{LO}(t) \\ &= A\cos(\phi_s - \phi_{\mathrm{LO}}) \\ &\quad + \cos(2\omega t + \phi_s + \phi_{\mathrm{LO}}) \end{aligned}$$

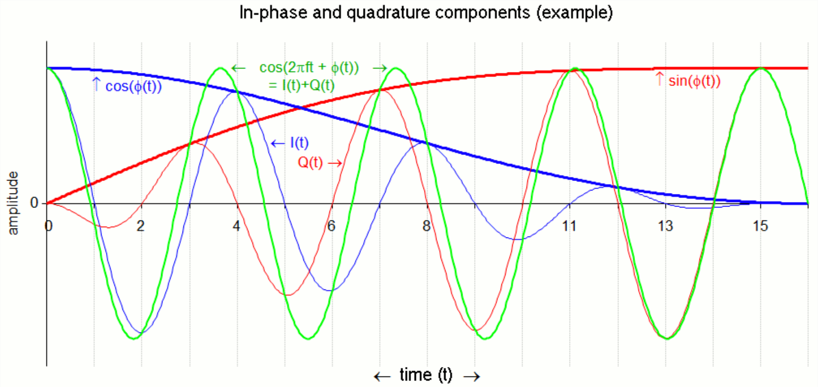
Pulses



a
Grover algorithm
State tomography
$|0\rangle$
$|0\rangle$
$R_y^{\pi/2}$
cU_{10}
cU_{00}
R_x^{π}
$R_x^{\pi/2}$
Readout
b
I_L
Q_L
I_R
4σ
Q_R
V_L
V_R
M
0
20
40
60
80
100
120
time (ns)

IQ modulation

$$M(t) = I(t)\cos(\omega t) + Q(t)\sin(\omega t) = A(t)\cos(\omega t + \phi(t))$$



Sources i

References

[1] L. DiCarlo et al. **"Demonstration of two-qubit algorithms with a superconducting quantum processor".** In: *Nature* 460.7252 (2009).